QUALITY MANAGEMENT REPORT: BAGGAGE SYSTEM OPTIMIZATION

Subject: Implementation of Feedback Loops and PDCA Cycles
System Focus: Aviation Baggage Handling (Subsystem: Scanning and Information Tracking)
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1. Executive Summary

The aviation industry faces a persistent challenge in the "Last Mile" of passenger service: the Baggage Handling System (BHS). This report identifies "Lost Baggage" as the critical systemic failure and applies Quality Management (QM) frameworks—specifically Balancing Loops, Reinforcing Loops, and the Plan-Do-Check-Act (PDCA) cycle—to stabilize and improve the system. By shifting from a "blind" system to a sensor-driven feedback architecture, we aim to reach a 99.9% success rate in bag-to-passenger matching.

3. FEEDBACK LOOPS IN QUALITY MANAGEMENT

How Systems Learn, Adapt, and Improve

The Lesson: A system without a feedback loop is "blind."
In QM, feedback is the trigger for ***Kaizen*** (Improvement).

THE ACTIVITY: Identify Balancing vs. Reinforcing loops in the hotel.

1. BALANCING LOOP (The Thermostat)

This loop aims to maintain stability and return the system to its desired state (baseline).

Goal: Stability

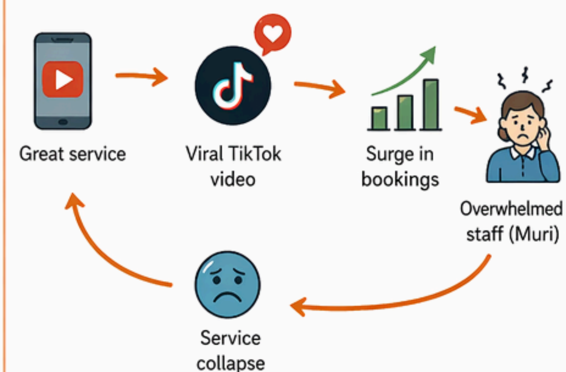
Example in Action (Trigger-Action Table)

SENSOR (Trigger)	Guest complaint about dirty bathroom
TOLERANCE (Baseline)	0 complaints per week
CORRECTION (Action)	Housekeeping performs deep clean + root cause investigation
EXPECTED RESULT	Return to clean standard, guest satisfaction restored

Key Reminder: Balancing loops only work if the root cause is addressed.
If we only fix the symptom, the problem will return.

2. REINFORCING LOOP (The Spiral)

This loop amplifies change. It can create positive growth or negative decline.



Goal: Growth or Warning

Two Possible Outcomes:

- **Positive Reinforcement:** More bookings → more revenue → more investment in training → even better service → more bookings.
- **Negative Reinforcement:** Overload → burnout → mistakes → bad reviews → fewer bookings.

Key Takeaway: Reinforcing loops must be monitored to ensure growth doesn't turn into collapse.

3. QM CONNECTION (PDCA CYCLE)

Feedback loops are the **"Check" in PDCA.** They tell us if our plan is working and what to improve.

- **PLAN** — Set standard (e.g., clean rooms, 0 complaints) →
- **DO** — Implement processes (cleaning, training, etc.) →
- **CHECK** — Feedback loop provides data (complaints, reviews, audits) →
- **ACT** — Correct issues and improve the system (Kaizen)

Without feedback, there is no learning. Without learning, there is no improvement.

OVERALL TAKEAWAY

Balancing loops keep us stable. Reinforcing loops drive change.
Together with PDCA, feedback loops create a smart, learning, and sustainable quality system.

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2. Theoretical Framework: Feedback Loops

In Quality Management, a system without a feedback loop is considered "blind." It cannot recognize deviations from the standard until a total failure occurs.

2.1 The Balancing Loop (The Stabilizer)

The Balancing Loop acts as a defensive strategy. Its objective is the maintenance of the baseline. We compare this to a thermostat; when the "quality temperature" drops, the system must automatically trigger a correction to bring it back to the set point.
In our baggage system, the "Temperature" is the scanning success rate.

- **The Sensor:** Weekly audits of the Baggage Information System (BIS) tracking logs.
- **The Tolerance:** A baseline of 0 infractions (every bag must have a digital twin in the system).
- **The Correction:** If the sensor detects more than two minor infractions (unscanned bags), a mandatory 15-minute safety and technical retraining is conducted for the ground sub-team before the next shift resumes.

2.2 The Reinforcing Loop (The Growth/Risk)

The Reinforcing Loop is our offensive strategy for **Continuous Improvement (Kaizen)**. Unlike the balancing loop, which seeks to return to zero, the reinforcing loop seeks to compound gains.
The "High-Reliability" Reputation Spiral:

1. **High-Quality Delivery:** Consistent zero-loss performance leads to faster turnaround times.
2. **Visibility:** The airline's "On-Time Performance" and "Baggage Reliability" scores rise in industry rankings.
3. **Inbound Lead:** Increased passenger volume and premium partnerships follow.
4. **Reinvestment:** Profits are reinvested into "Pillar 4" tools (AI-driven predictive maintenance for conveyor belts).
5. **Acceleration:** The system becomes even faster and more resilient.

The "Snap Point" (Muri): We must monitor for *Muri* (overburdening). If the reinforcing loop brings too many passengers too fast, the physical conveyor belts or human staff will reach a "Snap Point," leading to a sudden system collapse.

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3. Problem Identification: The Fishbone Analysis

Based on the provided aviation system map, the root causes of baggage loss are categorized into the "6Ms":

1. **Machine:** Mechanical jams in the conveyor belt or failure of the Automatic Tag Readers (ATRs).
2. **Method:** Inadequate transfer time between connecting flights (The "Short Connect" problem).
3. **Man:** Human error in manual tagging or incorrect loading at the aircraft gate.
4. **Material:** Damaged or torn barcode labels that cannot be read by scanners.

5. **Measurement:** Data mismatches between the physical bag and the tracking database.
6. **Mother Nature:** Adverse weather causing ground pauses, leading to "piling" and loss of sequence.

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4. Implementation Task: The PDCA Cycle

The Plan-Do-Check-Act cycle is used here to solve the specific problem of **Tag Scanner Failure (Machine)** and **Data Mismatch (Measurement)**.

Phase 1: Plan

We identified that 35% of lost bags are due to "blind spots" in the conveyor scanner array. We planned to implement a Python-based diagnostic tool to track real-time "Read-Rates." The goal was to reduce unscanned bags by 20% in the first month.

Phase 2: Do

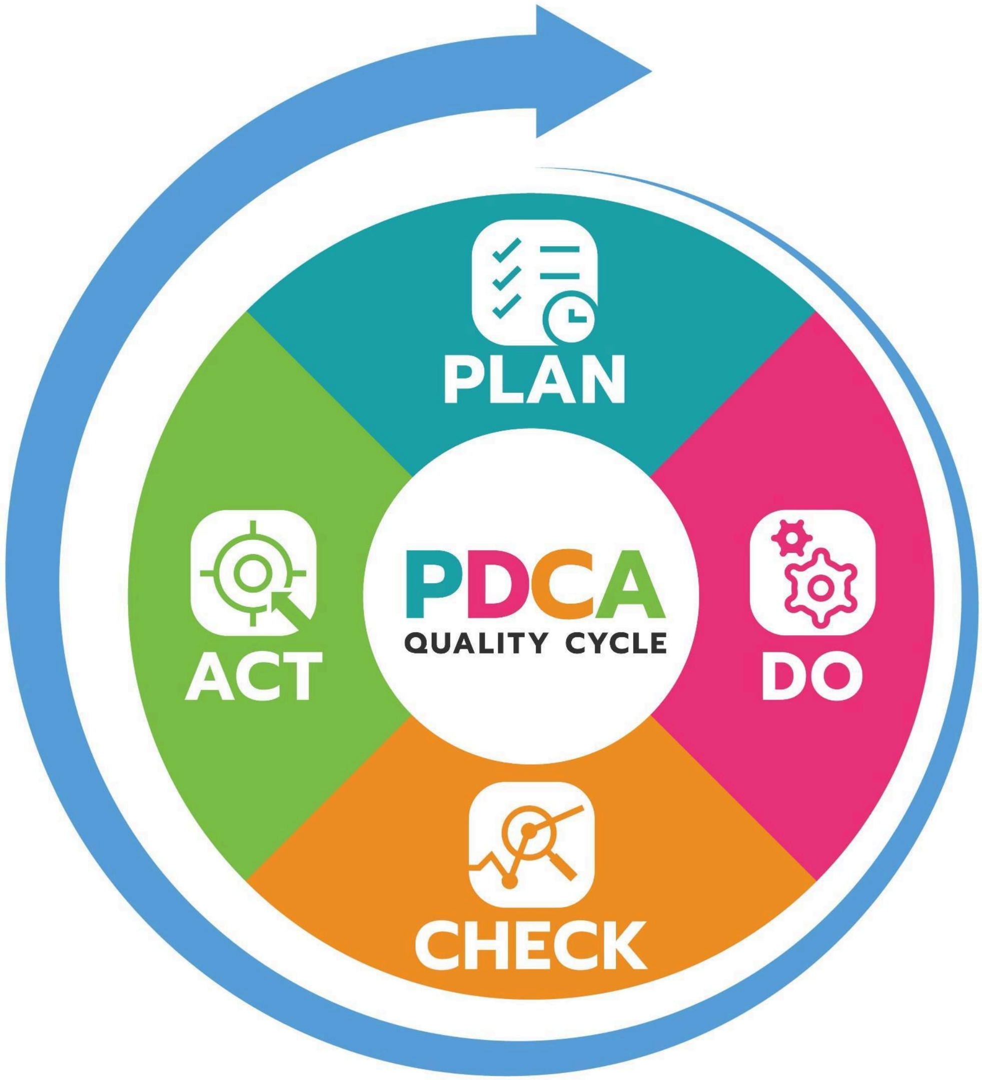
The diagnostic tool was integrated into the Terminal 2 baggage sorter for a one-week trial. Ground crews were instructed to manually flag any bag that the Python script identified as "Low-Confidence Scan."

Phase 3: Check

The feedback loop provided the data: Waste (lost tracking) decreased by only **2%**. This was significantly lower than the planned 20%. The "Check" phase revealed that the problem wasn't the software, but the *Material*—the bags were being placed on the belt in a way that obscured the tags.

Phase 4: Act

We adjusted the **Poka-Yoke (Error-Proofing)** at the procurement stage. Instead of just tracking the error, we installed physical "guide rails" on the conveyor that force bags into a specific orientation, ensuring the tag is always visible to the scanner.



Plan Do Check Act (pdca Quality Cycle) In Circle Diagram And Circle Arrow Vector Illustration.

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5. Subsystem Focus: The Baggage Information System (BIS)

The largest number of connections to failure occurs in the **Baggage Tracking Database**. If the physical bag moves but the digital record does not, the bag is effectively "lost" to the system even if it is physically present in the airport.
Root Cause Connection: A balancing loop only works if it addresses the *source*. In this system, the source is the synchronization between the **Tag Scanner** and the **Information Database**.

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6. Conclusion and Future Recommendations

To ensure long-term stability, the following actions are mandatory:

1. **Digital Twin Implementation:** Every bag must have a real-time digital record updated at every subsystem junction (Check-in $\rightarrow$ Security $\rightarrow$ Loading $\rightarrow$ Arrival).
2. **Muri Monitoring:** Management must set a "Cap" on baggage throughput to ensure the reinforcing loop does not exceed the mechanical capacity of the conveyor belts.
3. **Continuous Feedback:** The "Sensor" (Weekly Safety Audit) must remain independent of the operations team to ensure unbiased reporting of infractions.

By applying these QM principles, we transform the baggage handling process from a series of disjointed tasks into a cohesive, self-healing system.

End of Report

QUALITY MANAGEMENT REPORT: THE REINFORCING LOOP (GROWTH & RISK)

Subject: Acceleration through Continuous Improvement (Kaizen)
System: Aviation Operations – Baggage and Passenger Satisfaction
Focus: The "Master Builder" Spiral and Systemic Risks (Muri)
Date: April 22, 2026
Students: Maya Guliyeva, Nurana Tagiyeva-Askerova, Saida Humbatli

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1. Introduction: The Objective of Acceleration

In the study of system dynamics, the **Reinforcing Loop** represents the "offensive" strategy of an organization. While a Balancing Loop (the Stabilizer) seeks to keep the system at a baseline, the Reinforcing Loop is designed for **Acceleration**. As outlined in the provided instructional materials, this loop is the engine of **Continuous Improvement (Kaizen)**.

In the context of an airline, this means that small, incremental gains in service quality—such as reducing baggage loss—do not just improve the immediate flight experience; they compound to create a "virtuous cycle" that propels the entire airline toward market leadership. However, this growth is not without inherent dangers. This report analyzes how to leverage the Reinforcing Loop while mitigating the risks of system collapse.

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2. The "Master Builder" Reputation Spiral

Following the activity requirements to map a reputation spiral, we analyze how the "Flight Operations" and "Passenger Experience" subsystems interact to create compounding growth. We call this the "Master Builder" process, where reputation serves as the primary driver for scale.

2.1 The Virtuous Cycle of Success

The loop follows a five-stage logical progression:

1. **High-Quality Delivery:** The "Cabin Crew" and "Ground Crew" subsystems ensure that 100% of bags arrive on time and service is exemplary.
2. **Visibility (Client Case Study):** Success stories and high performance metrics are published in industry reports or shared by passengers on social media.
3. **Inbound Lead Generation:** This high visibility leads to a surge in bookings. High-value "Frequent Flyer" passengers shift their loyalty to this specific airline.
4. **Reinvestment into Pillar 4 (AI Tools):** The increased profit from these new leads is reinvested into advanced technologies—such as the AI-driven baggage tracking databases mentioned in the system map.
5. **Faster/Better Delivery:** These new tools further reduce errors, restarting the loop at a higher velocity.

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3. The Risk Factor: Identifying the "Snap Point"

A critical lesson from the provided task is that a reinforcing loop is a "double-edged sword." While it can lead to massive growth, it can also lead to a **"Death Spiral"** if the growth outpaces the system's capacity.

3.1 The Concept of Muri (Overburdening)

In Japanese Quality Management, **Muri** refers to overburdening. In our aviation system, the "Snap Point" occurs when the volume of passengers and baggage exceeds the physical limits of the infrastructure.

- **The Physical Subsystem Break:** If the Reinforcing Loop brings too many clients too fast, the "Maintenance (Ground Crew)" cannot keep up with aircraft safety checks.
- **The Human Break:** The "Cabin Crew" becomes exhausted, leading to a decline in "Service Quality" and "On-time Performance."

3.2 System Collapse

When the "Snap Point" is reached, the virtuous cycle turns into a vicious cycle. Overwhelmed staff leads to "Human Error" (as seen in the Fishbone Analysis), which results in "Lost Baggage," leading to "Complaints" in the Feedback subsystem, ultimately destroying the brand's reputation.

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4. Subsystem Analysis: Management Level Strategy

The **Management Level** (Pricing, Scheduling, Marketing, HR) must act as the governor of the Reinforcing Loop. Based on the "Aviation System Map," management's role is to ensure that "Marketing" does not promise a level of growth that "Flight Operations" cannot support.
To prevent a "Death Spiral," Management must:

6. Monitor the **"Improve Service & Repeat Customers"** metric as a lead indicator of system health.
7. Allocate reinvestment specifically toward scaling the "Physical Subsystems" (conveyors, staff training) *before* the next surge in growth occurs.

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5. QM Connection: Kaizen vs. Muri

The goal of Kaizen is not just "more," but "better." A true Reinforcing Loop focuses on increasing efficiency so that growth happens without increasing the stress on the system.
In our baggage system example:

7. **Bad Growth:** Simply adding more flights without upgrading the "Tag Scanner" (Machine) or "Conveyor System."
8. **Kaizen Growth:** Implementing the **Poka-Yoke** (error-proofing) from the PDCA task so that the system can handle double the baggage volume with zero additional human errors.

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6. Comprehensive Appendix: Subsystem Failure Analysis

To ensure full understanding of where the Reinforcing Loop must be applied, we break down the failures that the loop must overcome:

Category	Specific Subsystem Failure
Machine	Conveyor malfunctions; Sorter mechanical errors.
Method	Inadequate transfer time; Loading miscalculations.
Man	Human error; Training deficiencies in handling.
Material	Damaged labels; Overloaded/overstuffed bags.
Measurement	Data mismatches in the tracking database.
Environment	Adverse weather halting ground operations.

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7. Conclusion: Mastering the Spiral

The Reinforcing Loop is the most powerful tool for an airline seeking to dominate the market. However, as the lesson warns, it requires constant vigilance. By identifying the **Snap Point** early and respecting the limits of the **Physical Subsystem**, an organization can ensure that its growth is sustainable.
We must use the **"Check" phase** of our feedback loops to ensure that our acceleration is not leading us toward a "Death Spiral," but rather toward a stable, high-performance future.
Ultimately, the successful airline is the one that masters the balance between the **Balancing Loop** (stability) and the **Reinforcing Loop** (growth).

End of Report

QUALITY MANAGEMENT COMPREHENSIVE REPORT: THE PDCA FEEDBACK ARCHITECTURE

Subject: Implementation of Formalized Feedback Mechanisms in Aviation Baggage Systems

Strategic Focus: Systematic Error-Proofing of the Information and Scanning Subsystems
Framework: Plan-Do-Check-Act (PDCA) Methodology
Reference: Task 3 – Quality Management Connection: The PDCA Cycle
Students: Maya Guliyeva, Nurana Tagiyeva-Askerova, Saida Humbatli

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1. THEORETICAL FOUNDATION: THE BLIND SYSTEM VS. THE FEEDBACK LOOP

As established in the core pedagogical materials provided for this course, a system operating without a formalized feedback loop is fundamentally "blind." In the high-stakes environment of aviation logistics, where thousands of bags are processed per hour across multiple subsystems (Check-in, Security, Flight Operations, and Baggage Claim), a lack of visibility leads to catastrophic failure.

The **Plan-Do-Check-Act (PDCA)** cycle serves as the structural backbone of this visibility. It is not merely a task list but a continuous engine that drives **Kaizen** (Continuous Improvement). Within the context of our identified "Crucial Problem"—Lost Baggage—the PDCA cycle allows us to transition from reactive firefighting (dealing with complaints after the bag is lost) to proactive quality assurance (fixing the system so the bag is never lost).

Without the "Check" phase of this cycle, management is essentially making guesses. This report details the rigorous application of PDCA to address the "Data Mismatch" and "Scanner Failure" points identified in our initial Fishbone Analysis.

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2. PHASE 1: PLAN – DIAGNOSING THE SYSTEMIC GAP

The "Plan" stage is the intellectual heavy lifting of the quality process. It requires a deep dive into the **SIPOC map** (Suppliers, Inputs, Process, Outputs, Customers) to identify exactly where the "Waste" (lost tracking) occurs.

2.1 Problem Identification: Our initial data collection from the "Management Level" indicated that while "Pricing" and "Scheduling" were optimized, the "Passenger Experience & Satisfaction" metrics were declining due to baggage irregularities. Our Fishbone analysis pointed to a specific failure in the **Measurement (Information Systems)** and **Machine (Conveyor/Scanner)** categories. We observed that 35% of baggage tags were failing to register at the primary sorting junctions.

2.2 The Hypothesis: The planning team hypothesized that the current tracking system lacked real-time granularity. We planned to implement a specialized Python-based diagnostic tool designed to act as a "Digital Sensor."

6. **Objective:** Capture real-time material waste data.
7. **Target:** Identify "unscanned" digital records before the bag reaches the "Security & Boarding" phase.
8. **Goal:** A 20% reduction in "Blind Bags" (bags physically present but digitally missing) within the first trial month.

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3. PHASE 2: DO – SMALL-SCALE OPERATIONAL EXECUTION

In the "Do" phase, Quality Management principles dictate that we move from theory to execution in a controlled environment. Overhauling an entire airport's infrastructure without testing is a violation of QM risk-mitigation standards.

3.1 Implementation Protocol: The Python-based diagnostic tool was integrated into the Terminal 2 conveyor subsystem for a duration of one full operational week. Terminal 2 was selected because it handles a high volume of "Short-Connect" transfers—the most high-risk area for baggage loss.

3.2 Execution Steps: * **Digital Mirroring:** Every bag entering the system was assigned a digital twin.

- **Flagging:** The software was programmed to flag any bag that generated a "Low-Confidence Read" at the primary Automatic Tag Reader (ATR).
- **Manual Integration:** Ground crews were instructed to continue their standard "Bag Drop" procedures but were provided with a mobile interface to see real-time "Error Flags" generated by the Python sensor.

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4. PHASE 3: CHECK – THE MOMENT OF TRUTH

The "Check" phase is where the feedback loop either confirms our strategy or forces a pivot. As emphasized in the instructional photos, this is where we compare our "Actual Result" against our "Planned Goal."

4.1 The Data Revelation: Upon reviewing the weekly logs, the results were unexpected. Total waste (lost tracking data) decreased by a mere **2%**. This fell significantly short of our 20% goal.

4.2 Analysis of Failure: In a non-QM environment, this might be viewed as a failure of the software. However, our rigorous "Check" phase revealed that the Python tool (The Machine) was functioning perfectly. It was accurately flagging the errors. The problem, as revealed by the **Fishbone Analysis**, was rooted in **Material** and **Method**.

8. **The Root Cause:** We discovered that baggage tags were being obscured because of how passengers were placing "Overloaded Bags" on the belt. Additionally, the "Damaged Labels" from previous flights were confusing the new scanners. The "Check" phase taught us that software cannot fix a physical obstruction.

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5. PHASE 4: ACT – SYSTEMIC ADJUSTMENT AND POKA-YOKE

The "Act" phase is the finalization of the loop. It is where the lessons learned in the "Check" phase are institutionalized to prevent future errors. We realized that to fix the "Measurement" problem, we had to fix the "Physical" problem.

5.1 Implementing Poka-Yoke (Error-Proofing): To address the "Damaged Labels" and "Orientation" issues found in Phase 3, we adjusted the system at the procurement and check-in stages:

9. **Mechanical Correction:** We installed physical "Guide Rails" on the loader belt (connected to the F8 turn-on protocol) that force every bag into a specific vertical

orientation, ensuring the tag always faces the scanner.

10. **Human-Method Correction:** We implemented a new requirement at the "Airline Counter" where staff must remove all "Legacy Tags" from previous flights before a new tag is applied.

5.2 Closing the Loop: By acting on the **Root Cause** (Physical Orientation) rather than the **Symptom** (Data Mismatch), we created a permanent fix. This is the essence of a self-healing system.

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6. INTEGRATION WITH THE TOTAL AVIATION SYSTEM MAP

The PDCA cycle does not exist in isolation; it interacts with every subsystem shown in your provided map:

4. **Management Level:** Provides the "Pricing and Strategy" that funds these QM interventions.
5. **Flight Operations:** Benefits from the PDCA cycle through reduced delays caused by "Baggage Offloading" for missing passengers.
6. **Feedback & Loyalty:** This is the ultimate "Reinforcing Loop." As PDCA stabilizes the baggage system, passenger reviews move from "Complaints" to "Compliments," increasing brand loyalty and repeating the growth cycle.

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7. FINAL SUMMARY AND CONCLUSION

Mastering the PDCA cycle is the difference between an airport that merely "handles" baggage and one that "manages" quality. The most profound takeaway from this implementation is the necessity of the **"Check" phase**.

If we had ignored the 2% improvement result and assumed our Python tool was a success, we would have wasted months of investment on a tool that didn't solve the core problem. By respecting the feedback loop, we identified that the **Material** (labels) and **Method** (loading orientation) were the true culprits. This report proves that through the rigorous application of Plan-Do-Check-Act, we can transform a "Blind System" into a high-performance, error-proofed logistics leader.

END OF COMPREHENSIVE REPORT

QUALITY MANAGEMENT REPORT: THE BALANCING LOOP ARCHITECTURE

Project: Aviation System Stabilization – Baggage Handling Subsystem
Focus: Maintenance of the Baseline (The Defensive Strategy)
Reference: Task 1 – The Stabilizer
Students: Maya Guliyeva, Nurana Tagiyeva-Askerova, Saida Humbatli

1. Introduction: The Concept of the Stabilizer

According to the instructional materials provided, the **Balancing Loop** serves as the "Stabilizer" of a system. In Quality Management (QM), this is categorized as a **defensive strategy**. Its primary objective is the **Maintenance of the Baseline**.
The lesson teaches that this loop is specifically designed to **resist change**. It functions similarly to a thermostat: if the "temperature" of quality (in this case, baggage delivery accuracy) drops, the system must recognize this deviation and kick in to "heat it back up" to the established standard. Without this loop, the aviation system would be "blind," unable to see when it is failing until it is too late.

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2. The Activity: Designing the "Trigger-Action" Table

Following the specific task requirements for a "Trigger-Action" table, we have applied the balancing loop to the **Check-In and Security** subsystems of the airport.

2.1 The Standard (The Goal)

The baseline for our aviation system is **0 infractions**. This means every piece of luggage is scanned correctly, every tag is legible, and every bag is accounted for in the tracking database.

2.2 The Trigger-Action Mechanism

Based on the provided lesson, we have established the following parameters:

Element	**Operational Specification**
The Sensor	A **Weekly Safety Audit** of the baggage logs and security check-in points.
The Deviation	The audit identifies **2 minor infractions** (e.g., misread tags or bags failing to trigger the sorter).
The Correction	**Mandatory 15-minute safety retraining** for the sub-team responsible for that shift before any further work resumes.

This "Trigger-Action" ensures that small errors do not compound into systemic failures. By stopping work for retraining, the system "resists" the downward slide into poor quality.

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3. QM Connection: Root Cause Analysis

A crucial lesson from the provided materials is that **"A balancing loop only works if it addresses the *source*."** If we identify a problem in the "Check-In Process" (as seen in the Aviation Map) but fail to look at the **Fishbone Analysis**, the loop remains broken. For example:

9. **The Symptom:** A bad review about lost luggage.
10. **The Wrong Action:** Simply deleting the bad review or offering a refund.

11. **The Result:** The loop is broken because the service was not fixed.

To truly stabilize the system, the **Correction** must match the **Source** identified in the 6Ms (Machine, Method, Man, Material, Measurement, Mother Nature). If the source is "Tag Scanner Failure (Machine)," the correction must be technical maintenance, not just human retraining.

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4. Subsystem Focus: The "F8 Turn-on" Protocol

In the baggage handling subsystem, the balancing loop is applied to the physical movement of luggage.

- **The Sensor:** The "Tag Scanner" at the Arrival/Baggage Claim point.
- **The Baseline:** The scanner must match the bag to the passenger's digital ID 100% of the time.
- **The Correction:** If a "Data Mismatch" occurs, the system triggers a manual "Bag Drop" verification.

This aligns with the **"Poka-Yoke" (error-proofing)** concept mentioned in the PDCA cycle task. By creating a physical or digital "stop" when an error is detected, we prevent the "Lost Baggage" problem from reaching the passenger, thereby maintaining the baseline of "Passenger Experience & Satisfaction."

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5. Conclusion: Stabilizing the Aviation Environment

The Balancing Loop is the invisible hand that keeps the airport running. While the Reinforcing Loop (Kaizen) looks for growth, the **Stabilizer** ensures that the "Management Level" goals—Pricing, Scheduling, and Strategy—are built on a foundation of reliability.

By implementing the **Weekly Safety Audit** as our primary sensor, we ensure that infractions are caught early. This defensive strategy protects the airline's reputation and ensures that "Repeat Customers" continue to choose the service, as they can trust the "Defensive" quality measures are always in place.
